CORE PURE / POLAR COORDINATES

Polar Coordinates Basics

Mark scheme 10 questions · 97 marks

Questions

Question	Marks by part	Marks	Page
Question 1	(a) 3 (b) 2 (c) 3	8	4
Question 2	(a) 1 (b) 3 (c) 4	8	5
Question 3	(a) 1 (b) 6	7	6
Question 4	(a) 4 (b) 4	8	8
Question 5	(a) 4 (b) 4 (c) 3	11	9
Question 6	(i) 4 (ii) 3	7	11
Question 7	(a) 3 (b) 3 (c) 2 (d) 4	12	12
Question 8	(i) 2 (ii)(a) 4 (ii)(b) 3 (iii) 4	13	14
Question 9	(a) 2 (b) 6	8	16
Question 10	(a) 3 (b)(i) 3 (b)(ii) 5 (b)(iii) 4	15	17
	Total	**97**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

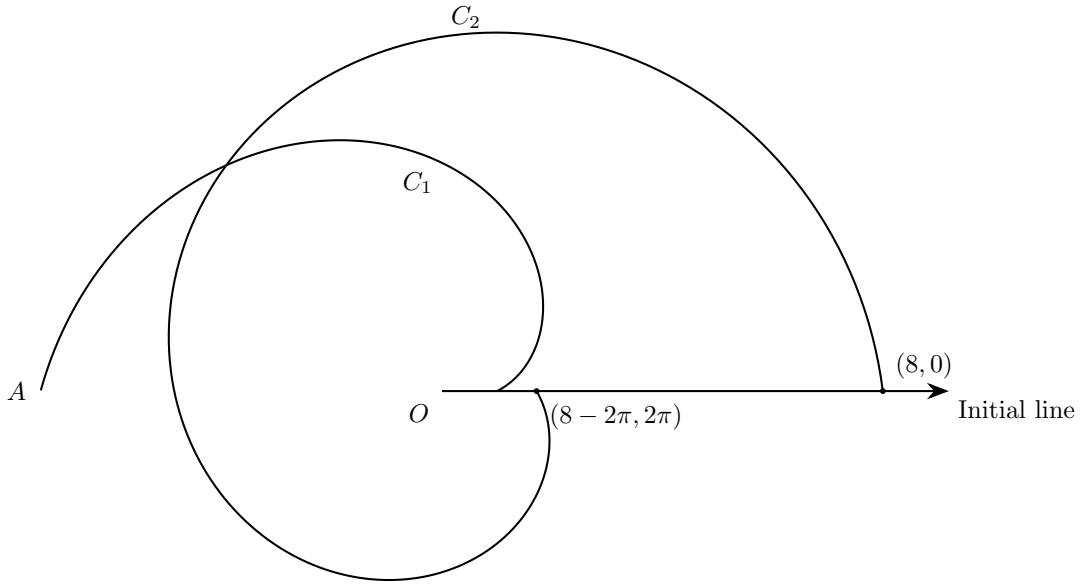
- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [***n***] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

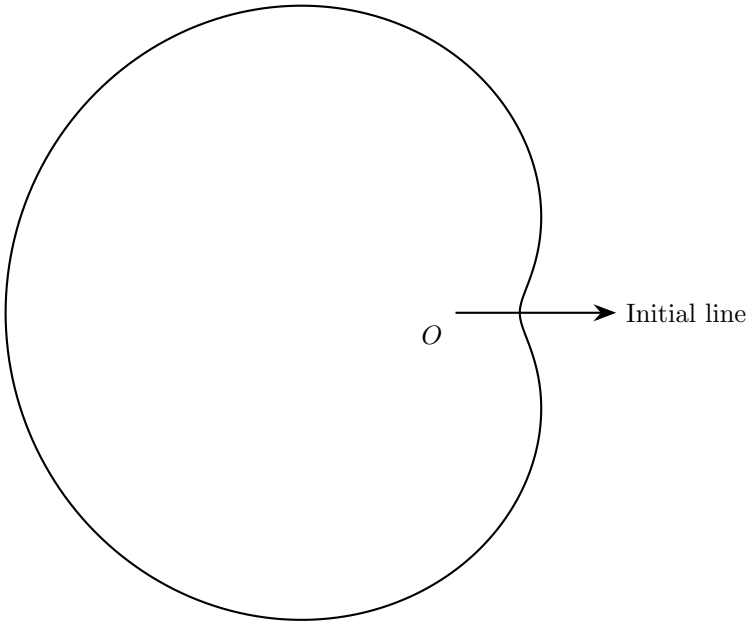
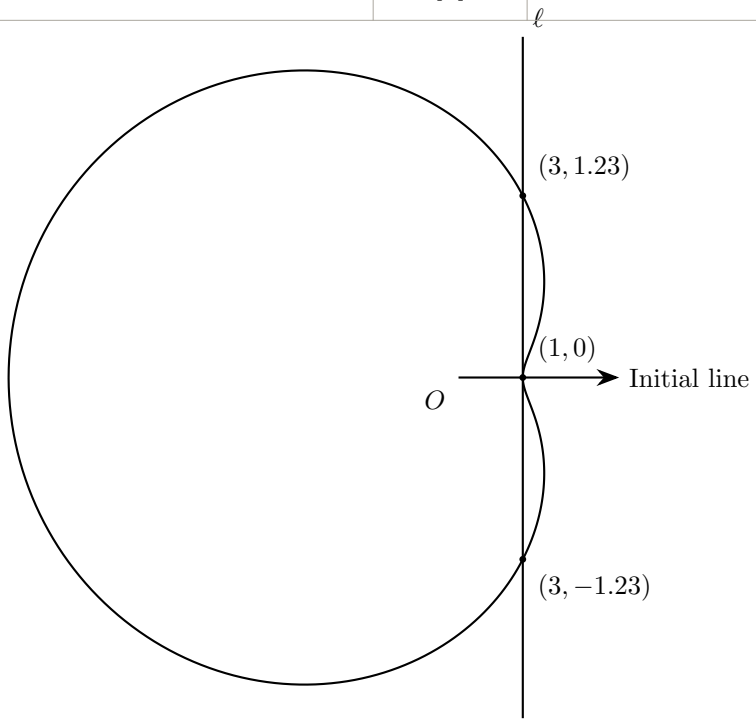
- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

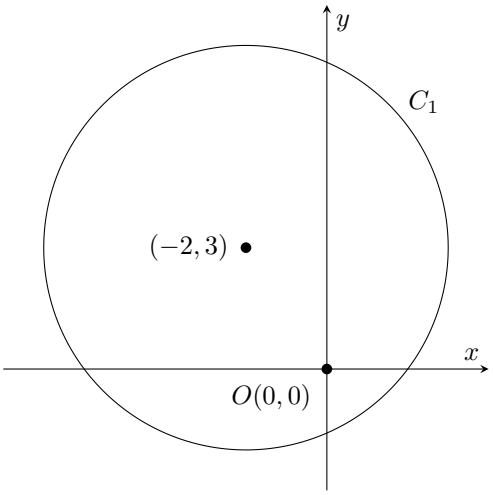
Question	Scheme	Marks	Notes
1(a)	$A = (1 + 2\pi, \pi)$	**B1**	Accept r awrt 7.28; $\theta = \pi$ or $180°$.
	$x = "(1 + 2\pi)" \cos \pi,\ \ y = "(1 + 2\pi)" \sin \pi$	**M1**	Uses $x = r\cos\theta$ and $y = r\sin\theta$ with their r and $\theta = \pi$.
	$(-1 - 2\pi, 0)$	**A1**	oe, or awrt $(-7.28, 0)$.
		[3]	
1(b)	$1 + 2\theta = 8 - \theta \Rightarrow \theta = \ldots$	**M1**	Equates the two expressions for r and solves for θ. Condone sign slips.
	$\left(\frac{17}{3}, \frac{7}{3}\right)$	**A1**	oe, or awrt $(5.67, 2.33)$. Both coordinates needed.
		[2]	
1(c)			
	Shape	**B1**	Spiral making one anticlockwise turn with r decreasing throughout, starting and ending on the initial line, not reaching O.
	Labelled point	**B1**	Start on the initial line labelled 8 or $(8, 0)$.
	Labelled point	**B1**	End on the initial line between O and the start, labelled $8 - 2\pi$, awrt 1.72, or $(8 - 2\pi, 2\pi)$.
		[3]	
	Total for Question 1	**8**	

Question	Scheme	Marks	Notes
2(a)	$r = \dfrac{6}{2-\sin\frac{\pi}{6}} = \dfrac{6}{2-\frac{1}{2}} = 4$	**B1**	Needs $\sin\frac{\pi}{6} = \frac{1}{2}$ substituted, or the denominator $\frac{3}{2}$ seen, leading to $r = 4$.
		[1]	
2(b)	r greatest when $2 - \sin\theta$ is least: $\sin\theta = 1$	**B1**	Identifies $\sin\theta = 1$, or $\theta = \frac{\pi}{2}$, as giving greatest r, by the least denominator or otherwise.
	$\theta = \frac{\pi}{2} \Rightarrow r = \dfrac{6}{2-1} = \ldots$	**M1**	Substitutes their θ, from maximising r, into the equation of C to find r.
	$\left(6, \frac{\pi}{2}\right)$	**A1**	Accept $r = 6$, $\theta = \frac{\pi}{2}$ stated separately.
		[3]	
2(c)	$r\sin\theta = r\cos\theta \Rightarrow \tan\theta = 1$	**M1**	Substitutes $x = r\cos\theta$, $y = r\sin\theta$ into $y = x$, reaching $\tan\theta = 1$ or an angle directly.
	$\theta = \frac{\pi}{4},\ -\frac{3\pi}{4}$	**A1**	Both, with no others in $-\pi < \theta \leq \pi$. Condone $\frac{5\pi}{4}$ for the second.
	$r = \dfrac{6}{2-\frac{\sqrt{2}}{2}},\ x = r\cos\frac{\pi}{4} = \ldots$ $r = \dfrac{6}{2+\frac{\sqrt{2}}{2}},\ x = r\cos\left(-\frac{3\pi}{4}\right) = \ldots$	**M1**	Finds r at one of their angles and uses $x = r\cos\theta$ or $y = r\sin\theta$. Need not be simplified.
	$\left(\frac{6+12\sqrt{2}}{7}, \frac{6+12\sqrt{2}}{7}\right), \left(\frac{6-12\sqrt{2}}{7}, \frac{6-12\sqrt{2}}{7}\right)$	**A1**	Both points, each with both coordinates, in any exact form; need not be rationalised.
		[4]	
Total for Question 2		**8**	

Question	Scheme	Marks	Notes
3(a)			
	Symmetry	**B1**	Given portion reflected in the initial line, giving a closed curve symmetric about it and still passing through $(1, 0)$.
		[1]	
3(b)			
	Intercepts	**B1**	Vertical line through $(1, 0)$ on the initial line, meeting C there and crossing it once above and once below the initial line.

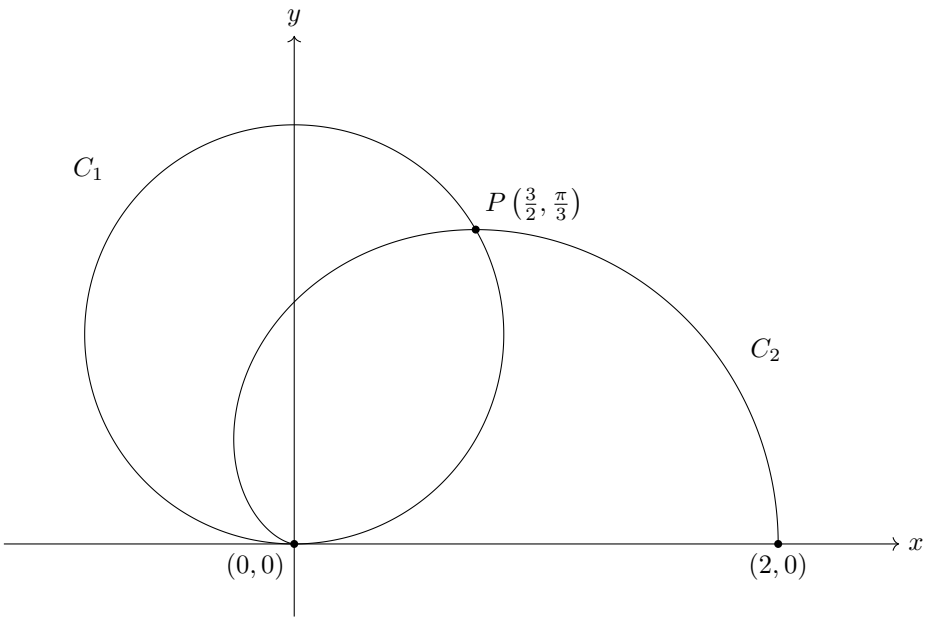
Question 3 continued

Question	Scheme	Marks	Notes
	$(4-3\cos\theta)\cos\theta = 1$	**M1**	Substitutes for r in $r\cos\theta = 1$ to obtain an equation in $\cos\theta$ alone.
	$3\cos^2\theta - 4\cos\theta + 1 = 0$	**A1**	oe, any three-term form.
	$(3\cos\theta - 1)(\cos\theta - 1) = 0 \Rightarrow \cos\theta = \ldots$ $\theta = \ldots,\ r = 4 - 3\cos\theta = \ldots$	**dM1**	Solves their three-term quadratic in $\cos\theta$ by any means, then finds at least one θ and its r from either equation.
	$(1, 0)$	**A1**	
	$(3, 1.23),\ (3, -1.23)$	**A1**	Both, with $r = 3$ and angles awrt ± 1.23. A0 for an angle outside $-\pi < \theta \leq \pi$.
		[6]	
	Total for Question 3	**7**	

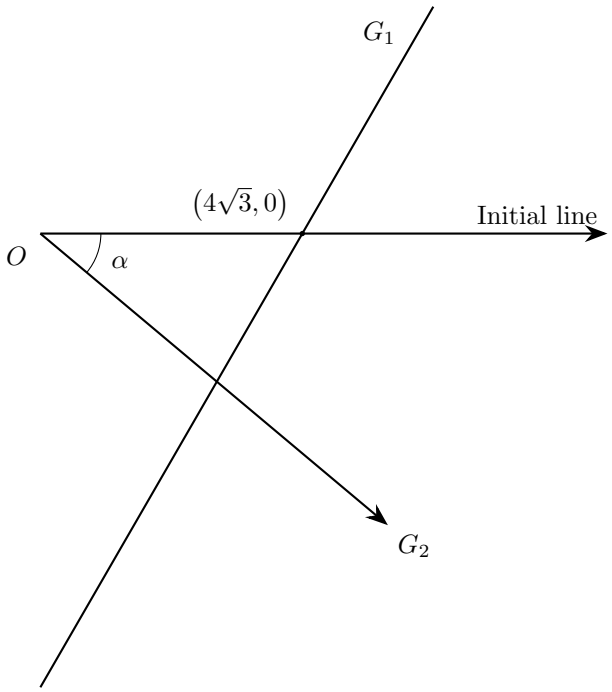
Question	Scheme	Marks	Notes
4(a)	$x^2 + y^2 + 4x - 6y = 12$	**M1 A1**	**M1** Replaces r^2 by $x^2 + y^2$ and at least one of $r\cos\theta$, $r\sin\theta$ by x, y. **A1** Correct equation, oe.
	$(x+2)^2 - 4 + (y-3)^2 - 9 = 12$	**M1**	Completes the square in both x and y for their equation of the form $x^2 + y^2 + ax + by = c$. Condone sign slips.
	$(x+2)^2 + (y-3)^2 = 25$ Centre $(-2, 3)$, radius 5	**A1**	Both centre and radius, from a correct equation in circle form.
		[4]	
4(b)			
	Centre and radius	**B1**	Circle with its centre marked in the second quadrant, labelled $(-2, 3)$.
	Labelled point	**B1**	Pole marked at the origin, inside the circle.
	$OA = \sqrt{(-2)^2 + 3^2}$ $[= \sqrt{13}]$ $5 \pm$ "$\sqrt{13}$"	**M1**	Finds the distance from the pole to their centre and adds it to and subtracts it from their radius. No justification that the pole is inside is needed.
	Greatest $5 + \sqrt{13}$, least $5 - \sqrt{13}$	**A1**	Both, correctly identified.
		[4]	
	Total for Question 4	**8**	

Question	Scheme	Marks	Notes
5(a)	$x^2 + \left(y - \frac{\sqrt{3}}{2}\right)^2 = \ldots$	**M1**	Completes the square in y, or compares with the general circle equation. Condone a slip in the constant.
	Centre $\left(0, \frac{\sqrt{3}}{2}\right)$, radius $\frac{\sqrt{3}}{2}$	**A1**	Both, exact or awrt 0.866.
	$r^2\cos^2\theta + r^2\sin^2\theta = \sqrt{3}\,r\sin\theta$	**M1**	Substitutes $x = r\cos\theta$, $y = r\sin\theta$, or uses $x^2 + y^2 = r^2$ with $y = r\sin\theta$.
	$r^2 = \sqrt{3}\,r\sin\theta$, $r \neq 0$ $\Rightarrow r = \sqrt{3}\sin\theta$	**A1 AG**	Correct substituted equation leading without error to the printed result. Needs $r \neq 0$, or non-pole, stated when dividing by r.
		[4]	
5(b)	$\sqrt{3}\sin\theta = 1 + \cos\theta$ $\Rightarrow \sqrt{3}\sin\theta - \cos\theta = R\sin(\theta - \alpha) = 1$	**M1**	Equates the radii and reduces to one trigonometric function, e.g. any harmonic form with R and α attempted, or half-angle identities. Condone sign slips.
	$R = 2,\ \alpha = \frac{\pi}{6}$	**A1**	Or a correct equivalent single-function equation by another route.
	$\sin\left(\theta - \frac{\pi}{6}\right) = \frac{1}{2} \Rightarrow \theta = \ldots$	**dM1**	Solves their equation for θ in $0 \leq \theta \leq \pi$, finding at least one value.
	$\theta = \frac{\pi}{3}$ ($\theta = \pi$ is the pole) $P\left(\frac{3}{2}, \frac{\pi}{3}\right)$	**A1**	cso. Exact. Needs $\theta = \pi$ rejected, or $\frac{\pi}{3}$ shown to be the only solution in $0 < \theta < \pi$, with one intersection stated.
		[4]	

Question 5 continued

Question	Scheme	Marks	Notes
5(c)			
	Shape	**B1**	Complete circle above the initial line, touching it at the pole.
	Shape	**B1**	Upper half of a cardioid from $(2,0)$, passing into $x < 0$ before reaching the pole; both curves labelled.
	Intercepts	**B1**	P marked at the intersection, and $(2,0)$ and $(0,0)$ indicated.
		[3]	
	Total for Question 5	**11**	

Question	Scheme	Marks	Notes
6(i)	$r^4 = 5r^2\cos^2\theta + 4r^2\cos\theta\sin\theta + 2r^2\sin^2\theta$	**M1**	Substitutes $x = r\cos\theta$, $y = r\sin\theta$ and $x^2 + y^2 = r^2$ throughout. Condone one slip.
	$r^2 = 5\cos^2\theta + 4\cos\theta\sin\theta + 2\sin^2\theta$	**A1**	
	$r^2 = \frac{5}{2}(1 + \cos 2\theta) + 2\sin 2\theta + (1 - \cos 2\theta)$	**M1**	Uses double-angle identities for all of $\cos^2\theta$, $\sin^2\theta$ and $\sin\theta\cos\theta$. Condone sign and coefficient slips.
	$r^2 = \dfrac{7 + 3\cos 2\theta + 4\sin 2\theta}{2}$	**A1 AG**	Correct equation in r and θ leading without error to the printed result; combined steps accepted.
		[4]	
6(ii)	$3\cos 2\theta + 4\sin 2\theta = R\cos(2\theta - \alpha)$ $\dfrac{7 - R}{2} \leq r^2 \leq \dfrac{7 + R}{2}$	**M1**	Expresses $3\cos 2\theta + 4\sin 2\theta$ as any single harmonic form and uses its bounds $\pm R$ in r^2. Condone a wrong R.
	$R = 5 \Rightarrow 1 \leq r^2 \leq 6$	**A1**	
	Least distance 1, greatest distance $\sqrt{6}$	**A1**	Both distances, exact. Not for the values of r^2 alone.
		[3]	
	Total for Question 6	**7**	

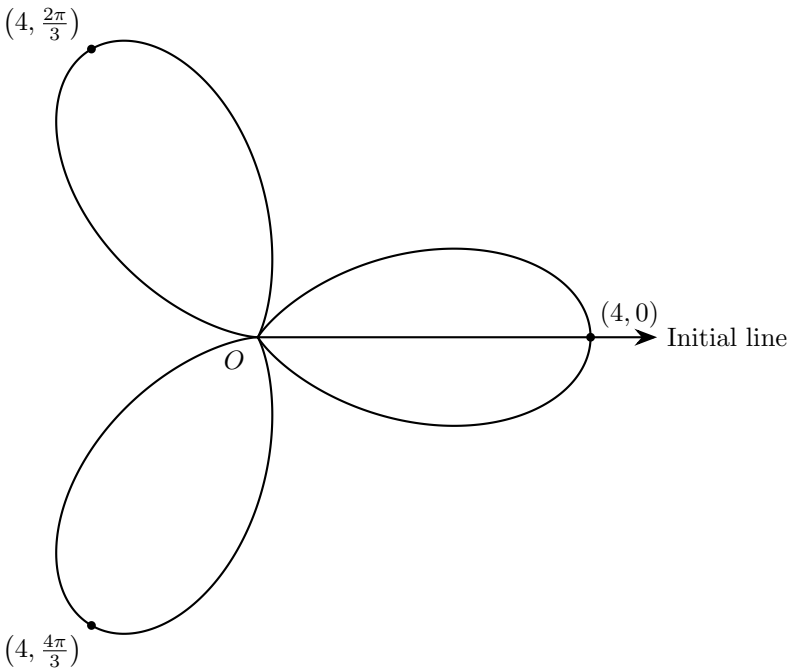
Question	Scheme	Marks	Notes
7(a)	$r\cos\left(\theta+\frac{\pi}{6}\right)=6$ $r\left(\cos\theta\cos\frac{\pi}{6}-\sin\theta\sin\frac{\pi}{6}\right)=6$	**M1**	Multiplies through by $\cos\left(\theta+\frac{\pi}{6}\right)$ and expands the compound angle. Condone sign slips in the expansion.
	$\frac{\sqrt{3}}{2}x-\frac{1}{2}y=6$	**dM1**	Replaces $r\cos\theta$ by x and $r\sin\theta$ by y.
	$y=\sqrt{3}x-12$	**A1**	cso. Must be in the form $y=mx+c$; another linear form such as $\sqrt{3}x-y=12$ scores A0.
		[3]	
7(b)			
	Shape	**B1**	Straight line with positive gradient crossing the initial line to the right of O, extending above and below it.
	Half-line	**B1**	Ray from O below the initial line meeting G_1, with α marked between the ray and the initial line.
	Labelled point	**B1**	$\left(4\sqrt{3},0\right)$, $\left(6\sec\frac{\pi}{6},0\right)$ or awrt $(6.93,0)$, at the crossing of G_1 and the initial line.
		[3]	
7(c)	$r=6\sec\left(-\alpha+\frac{\pi}{6}\right)$	**M1**	Substitutes $\theta=-\alpha$ into the equation of G_1.

Question 7 continued

Question	Scheme	Marks	Notes
	$\left(6\sec\left(\frac{\pi}{6}-\alpha\right),\ -\alpha\right)$	**A1**	Both coordinates, oe such as $\frac{6}{\cos\left(\frac{\pi}{6}-\alpha\right)}$ or expanded.
		[2]	
7(d)	$12\sqrt{3}=\frac{1}{2}\left("4\sqrt{3}"\right)\left("6\sec\left(\frac{\pi}{6}-\alpha\right)"\right)\sin\alpha$	**M1 A1**	**M1** Forms the triangle's area in terms of α, e.g. $\frac{1}{2}ab\sin C$ with their sides from O and angle α, or $\frac{1}{2}\int r^2\,\mathrm{d}\theta$, equal to $12\sqrt{3}$. **A1** Correct equation, oe.
	$\sin\alpha=\cos\left(\frac{\pi}{6}-\alpha\right)$ $\quad=\frac{\sqrt{3}}{2}\cos\alpha+\frac{1}{2}\sin\alpha$ $\Rightarrow\tan\alpha=\ldots$	**dM1**	Expands the compound angle and rearranges to an equation in $\tan\alpha$ alone. Condone sign slips.
	$\tan\alpha=\sqrt{3}\Rightarrow\alpha=\frac{\pi}{3}$	**A1**	cso, with no other values given.
		[4]	
	Total for Question 7	**12**	

Question	Scheme	Marks	Notes
8(i)	$2(1+\cos 3\theta)=0 \Rightarrow \cos 3\theta=-1$ $\Rightarrow \theta=\ldots$	**M1**	Sets $r=0$ and solves $\cos 3\theta=-1$ for at least one value of θ.
	$\alpha=\frac{\pi}{3}$	**A1**	In terms of π; 1.05 alone scores A0.
		[2]	
8(ii)(a)	$f\left(\frac{2k\pi}{3}-\theta\right)=2[1+\cos(2k\pi-3\theta)]$ $=2[1+\cos 2k\pi\cos 3\theta+\sin 2k\pi\sin 3\theta]$	**M1**	Expands $\cos(2k\pi-3\theta)$ by the addition formula, or uses the evenness and period 2π of cosine.
	$=2(1+\cos 3\theta)=f(\theta)$	**A1**	Reaches $f(\theta)$ without error for general integer k, not a particular value.
	$f\left(\theta+\frac{2\pi}{3}\right)=2[1+\cos(3\theta+2\pi)]$ $=2[1+\cos 3\theta\cos 2\pi-\sin 3\theta\sin 2\pi]$	**M1**	Expands $\cos(3\theta+2\pi)$ by the addition formula, or uses the period 2π of cosine.
	$=2(1+\cos 3\theta)=f(\theta)$	**A1 AG**	Correct substituted expression leading without error to $f(\theta)$, with $=f(\theta)$ or a statement in words to conclude.
		[4]	
8(ii)(b)	Reflection in $\theta=\beta$: $\theta\to 2\beta-\theta$ $2\beta=\frac{2k\pi}{3}\Rightarrow\theta=\frac{k\pi}{3}$	**B1**	Lines $\theta=\frac{k\pi}{3}$ oe, linked to the first identity.
	$\theta=0,\ \frac{\pi}{3},\ \frac{2\pi}{3}$	**B1**	All three and no others.
	Rotation through $\frac{2\pi}{3}$	**B1**	
		[3]	

Question 8 continued

Question	Scheme	Marks	Notes
8(iii)			
	Shape	**B1**	Three equal closed lobes meeting at O, one centred on the initial line, the others in roughly the correct directions.
	$r_{\max} = 4$	**B1**	
	$\cos 3\theta = 1 \Rightarrow 3\theta = 2k\pi \Rightarrow \theta = \ldots$	**M1**	Sets $\cos 3\theta = 1$ and finds at least one nonzero value of θ.
	$\theta = 0,\ \frac{2\pi}{3},\ \frac{4\pi}{3}$	**A1**	All three, or 0, awrt 2.09 and 4.19, with no others in $0 \leq \theta < 2\pi$.
		[4]	
	Total for Question 8	**13**	

Question	Scheme	Marks	Notes
9(a)	$r\cos\left(\theta-\frac{\pi}{4}\right)=3$ $\Rightarrow \frac{r\cos\theta+r\sin\theta}{\sqrt{2}}=3$	**M1**	Expands $\cos\left(\theta-\frac{\pi}{4}\right)$ and replaces $r\cos\theta$, $r\sin\theta$ by x, y. Condone a slip in the value of $\cos\frac{\pi}{4}$.
	$x+y=3\sqrt{2}$	**A1 AG**	Correct expansion with $\cos\frac{\pi}{4}=\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}$, leading without error to the printed equation.
		[2]	
9(b)	$\sin 2\theta = 2\cos^2\left(\theta-\frac{\pi}{4}\right)-1$ $\quad = \frac{18}{r^2}-1$ $r=5-2\left(\frac{18}{r^2}-1\right)$	**M1**	Eliminates θ using L and a double-angle identity, e.g. via $(x+y)^2=r^2(1+\sin 2\theta)$, to an equation in r alone. Condone sign slips.
	$r^3-7r^2+36=0$	**A1**	oe, e.g. $r=7-\frac{36}{r^2}$.
	$(r-3)(r-6)(r+2)=0\Rightarrow r=\ldots$	**dM1**	Solves their cubic by any means, calculator included, to obtain values of r.
	$r=3,\ r=6$ ($r=-2$ rejected as $3\le r\le 7$)	**A1**	Both values, with $r=-2$ rejected for any valid reason.
	$\cos\left(\theta-\frac{\pi}{4}\right)=\frac{3}{\text{"}r\text{"}}$, $-\frac{\pi}{2}<\theta-\frac{\pi}{4}<\frac{\pi}{2}$ $\Rightarrow\theta=\ldots$	**M1**	Substitutes a positive value of their r into either equation and solves for θ in $-\frac{\pi}{4}<\theta<\frac{3\pi}{4}$.
	$\left(3,\frac{\pi}{4}\right),\ \left(6,-\frac{\pi}{12}\right),\ \left(6,\frac{7\pi}{12}\right)$ No other solutions in the range of L	**A1**	cso. All three exact points and no others, with a statement that the cosine equations have no further solutions in the range of L.
		[6]	
	Total for Question 9	**8**	

Question	Scheme	Marks	Notes
10(a)	$(r\cos\theta)^2 - (r\sin\theta)^2 = 6r\cos\theta$	**M1**	Substitutes both into the Cartesian equation.
	$r^2\cos 2\theta = 6r\cos\theta \Rightarrow r\cos 2\theta = 6\cos\theta$	**A1**	
	$x > 0 \Rightarrow \cos\theta > 0$, $r > 0 \Rightarrow \cos 2\theta > 0$ $\Rightarrow -\frac{\pi}{4} < \theta < \frac{\pi}{4}$, $r = 6\cos\theta\sec 2\theta$	**A1 AG**	Printed equation reached without error, with the range deduced from $\cos\theta > 0$ and $\cos 2\theta > 0$. A0 if the range is only stated.
		[3]	
10(b)(i)	$0 < \cos 2\theta = \cos^2\theta - \sin^2\theta \leq \cos^2\theta \leq \cos\theta$	**M1**	Attempts a valid justification: bounds $\cos 2\theta$ by $\cos\theta$ on the interval, sets $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 0$, or writes $r^2 = 2x^2 - 6x$ for $x \geq 6$.
	$r = \dfrac{6\cos\theta}{\cos 2\theta} \geq 6$, equality only at $\theta = 0$	**A1**	Or an equivalent conclusion: the stationary point shown to be a minimum, or r^2 increasing for $x \geq 6$.
	$N = (6, 0)$	**A1**	
		[3]	
10(b)(ii)	$6\sqrt{3}\cos 2\theta = 6\cos\theta$ $\Rightarrow \sqrt{3}(2\cos^2\theta - 1) = \cos\theta$	**M1**	Equates $6\sqrt{3}$ to the polar equation and uses a double-angle formula to reach an equation in $\cos\theta$ only; or eliminates y using $x^2 + y^2 = 108$.
	$2\sqrt{3}\cos^2\theta - \cos\theta - \sqrt{3} = 0$	**A1**	oe, or $2x^2 - 6x - 108 = 0$ oe.
	$(2\cos\theta - \sqrt{3})(\sqrt{3}\cos\theta + 1) = 0 \Rightarrow \cos\theta = \ldots$	**dM1**	Solves their three-term quadratic by factorising (product of constants correct), the formula or completing the square; calculator allowed.
	$\cos\theta = \frac{\sqrt{3}}{2}$ ($-\frac{1}{\sqrt{3}}$ rejected as $\cos\theta > 0$)	**A1**	Other root rejected with a reason, eg $\cos\theta > 0$ on the interval; or $x = 9$ with $x = -6$ rejected.
	$P\left(6\sqrt{3}, \frac{\pi}{6}\right)$, $Q\left(6\sqrt{3}, -\frac{\pi}{6}\right)$	**A1**	Both points exact, angles in the stated interval; $\frac{11\pi}{6}$ for Q scores A0.
		[5]	
10(b)(iii)	$NP^2 = 6^2 + (6\sqrt{3})^2 - 2(6)(6\sqrt{3})\cos\frac{\pi}{6}$	**M1**	Uses the cosine rule in triangle ONP with $ON = 6$, their OP from (b)(ii) and angle $\frac{\pi}{6}$, or the distance formula, or finds the Cartesian coordinates of P.

Question 10 continued

Question	Scheme	Marks	Notes
	$NP = 6$	**A1**	$NP = 6$, or $P(9, 3\sqrt{3})$ oe.
	$\angle ONP = \pi - 2\left(\frac{\pi}{6}\right)$ $[= \frac{2\pi}{3}]$ $\angle PNQ = 2\pi - 2\angle ONP = \ldots$	**M1**	Uses $ON = NP$ to find $\angle ONP$ and symmetry about the initial line to reach $\angle PNQ$; or any valid route, eg the direction of NP.
	$\angle PNQ = \frac{2\pi}{3}$	**A1**	Exact, in radians; 120° scores A0.
		[4]	
	Total for Question 10	**15**	